MAT41-2

Advanced Estimation

Continuous models, estimator comparison, and asymptotics

Method of moments and maximum likelihood for continuous distributions, and the asymptotic behaviour of estimators.

Albert School — 26 hours

Contents

1 Setup 2
2 Method of moments 2
3 Maximum likelihood for continuous densities 2
4 Comparing estimators 3
5 Convergence and consistency 4
6 Asymptotic normality of the MLE 4
7 The Cramér–Rao bound and efficiency 5
8 Asymptotic confidence intervals 5
8.1 Sample size determination 6

1 Setup

Throughout, $X_1, \ldots, X_n$ are independent and identically distributed (i.i.d.) random variables drawn from a continuous distribution with density $f(x; \theta)$ depending on an unknown parameter $\theta \in \Theta \subseteq \mathbb{R}^d$. An **estimator** of θ is a function $\hat{\theta} = \hat{\theta}(X_1, \ldots, X_n)$ of the sample. We use continuous random variables, densities, expectation and variance (MAT31-2) and the notions of estimator, bias and MLE for discrete models (MAT21-2).

The ***k*-th sample moment** is

$$m_k = \frac{1}{n} \sum_{i=1}^{n} X_i^k,$$

and the ***k*-th theoretical moment** is $\mu_k(\theta) = \mathbb{E}_\theta[X^k]$.

2 Method of moments

Definition 1 (Method-of-moments estimator) For a model with d parameters $\theta = (\theta_1, \ldots, \theta_d)$, the **method-of-moments** estimator is obtained by equating the first d theoretical moments to the corresponding sample moments and solving the system

$$\mu_k(\theta) = m_k, \quad k = 1, \ldots, d,$$

for θ. The solution $\hat{\theta}_{\mathrm{MM}}$ is the method-of-moments estimator.

For a one-parameter model ($d = 1$) this is the single equation $\mathbb{E}_\theta[X] = \frac{1}{n} \sum_{i=1}^{n} X_i$. For multi-parameter models one uses as many moment equations as there are parameters; if a moment equation is degenerate, the next higher moment (e.g. the second) is added, and the system is solved simultaneously.

3 Maximum likelihood for continuous densities

Definition 2 (Likelihood and log-likelihood) Given observations $x_1, \ldots, x_n$, the **likelihood** is

$$L(\theta) = \prod_{i=1}^{n} f(x_i; \theta),$$

and the **log-likelihood** is

$$\ell(\theta) = \log L(\theta) = \sum_{i=1}^{n} \log f(x_i; \theta).$$

Definition 3 (Maximum likelihood estimator) The **maximum likelihood estimator** (MLE) is

$$\hat{\theta}_{\mathrm{ML}} = \operatorname*{argmax}_{\theta \in \Theta} \ell(\theta).$$

Definition 4 (Score equation) The **score** is the derivative of the log-likelihood with respect to the parameter,

$$s(\theta) = \frac{\partial \ell}{\partial \theta} = \sum_{i=1}^{n} \frac{\partial}{\partial \theta} \log f(x_i; \theta).$$

An interior maximizer satisfies the **score equation** $s(\theta) = 0$.

Procedure (one parameter). To compute the MLE:
1. write the log-likelihood $\ell(\theta) = \sum_i \log f(x_i; \theta)$;
2. form the score $s(\theta) = \ell'(\theta)$ and solve the score equation $s(\theta) = 0$ for θ;
3. verify the **second-order condition** $\ell''(\theta) < 0$ at the solution, so it is a maximum (and not a minimum or saddle point), and check the boundary of Θ if the maximum may be attained there.

For a d-parameter model the score is the gradient $\nabla \ell(\theta) \in \mathbb{R}^d$, the score equation is the system $\nabla \ell(\theta) = 0$, and the second-order condition is that the Hessian $\nabla^2 \ell(\theta)$ be negative definite at the solution.

4 Comparing estimators

Let $\hat{\theta}$ be an estimator of a scalar parameter θ.

Definition 5 (Bias and mean squared error) The **bias** of $\hat{\theta}$ is

$$\operatorname{bias}\left(\hat{\theta}\right) = \mathbb{E}_\theta\left[\hat{\theta}\right] - \theta,$$

and its **mean squared error** (MSE) is

$$\operatorname{MSE}\left(\hat{\theta}\right) = \mathbb{E}_\theta\left[\left(\hat{\theta} - \theta\right)^2\right].$$

The estimator is **unbiased** if $\operatorname{bias}\left(\hat{\theta}\right) = 0$.

Proposition 6 (Bias-variance decomposition)

$$\operatorname{MSE}\left(\hat{\theta}\right) = \operatorname{Var}_\theta\left(\hat{\theta}\right) + \operatorname{bias}\left(\hat{\theta}\right)^2.$$

The **bias-variance tradeoff** is the principle, read off this decomposition, that estimation quality (small MSE) requires controlling both variance and squared bias, and that reducing one may increase the other; the best estimator minimizes their sum, not either term alone.

Definition 7 (Efficiency) Given two unbiased estimators $\hat{\theta}_1, \hat{\theta}_2$ of θ, $\hat{\theta}_1$ is **more efficient** than $\hat{\theta}_2$ if

$$\mathrm{Var}_\theta\left(\hat{\theta}_1\right) \leq \mathrm{Var}_\theta\left(\hat{\theta}_2\right)$$

for all θ, with strict inequality for some θ. The **relative efficiency** of $\hat{\theta}_1$ with respect to $\hat{\theta}_2$ is the ratio $\mathrm{Var}_\theta\left(\hat{\theta}_2\right)/\mathrm{Var}_\theta\left(\hat{\theta}_1\right)$.

To compare two estimators of the same parameter, compute the bias and MSE of each; among unbiased estimators, the one of smaller variance is more efficient.

5 Convergence and consistency

Let $(Y_n)_{n\geq 1}$ be a sequence of random variables and Y a random variable.

Definition 8 (Convergence in probability) Y_n **converges in probability** to Y, written $Y_n \xrightarrow{P} Y$, if for every $\varepsilon > 0$,

$$\lim_{n\to\infty} \mathbb{P}(|Y_n - Y| > \varepsilon) = 0.$$

Definition 9 (Almost sure convergence) Y_n **converges almost surely** to Y, written $Y_n \xrightarrow{\text{a.s.}} Y$, if

$$\mathbb{P}\left(\lim_{n\to\infty} Y_n = Y\right) = 1.$$

Proposition 10 (Relation between the two modes) Almost sure convergence implies convergence in probability; the converse does not hold in general.

The two modes differ in what is asserted: convergence in probability controls, for each large n, the probability of a deviation; almost sure convergence asserts that the random sequence itself has limit Y with probability one.

Definition 11 (Consistency) An estimator $\hat{\theta}_n$ (based on a sample of size n) is **consistent** for θ if $\hat{\theta}_n \xrightarrow{P} \theta$ as $n \to \infty$, for every $\theta \in \Theta$.

To verify consistency of a given estimator it suffices, for instance, to show $\mathrm{MSE}\left(\hat{\theta}_n\right) \to 0$, since this implies $\hat{\theta}_n \xrightarrow{P} \theta$.

6 Asymptotic normality of the MLE

Definition 12 (Fisher information) For a one-parameter model with density $f(x;\theta)$, the **Fisher information** (per observation) is

$$I(\theta)=\mathbb{E}_\theta\left[\left(\frac{\partial}{\partial\theta}\log f(X;\theta)\right)^2\right]=-\mathbb{E}_\theta\left[\frac{\partial^2}{\partial\theta^2}\log f(X;\theta)\right],$$

the two expressions being equal under the usual regularity conditions.

Theorem 13 (Asymptotic normality of the MLE) Under regularity conditions, the MLE $\hat{\theta}_n$ from an i.i.d. sample of size n is consistent and satisfies

$$\sqrt{n}\left(\hat{\theta}_n-\theta\right)\xrightarrow{d}\mathcal{N}\left(0,\frac{1}{I(\theta)}\right)$$

as $n\to\infty$. Equivalently, for large n, $\hat{\theta}_n$ is approximately $\mathcal{N}(\theta,1/(nI(\theta)))$.

The **asymptotic variance** of $\hat{\theta}_n$ is thus $1/(nI(\theta))$: the Fisher information $I(\theta)$ is the inverse of the (rescaled) asymptotic variance. Larger Fisher information means a smaller asymptotic variance, hence a more precise estimator.

7 The Cramér–Rao bound and efficiency

Theorem 14 (Cramér–Rao lower bound) Under regularity conditions, any unbiased estimator $\hat{\theta}$ of θ based on an i.i.d. sample of size n satisfies

$$\operatorname{Var}_\theta\left(\hat{\theta}\right)\geq\frac{1}{nI(\theta)}.$$

Definition 15 (Efficient estimator) An unbiased estimator is **efficient** if its variance attains the Cramér–Rao lower bound $1/(nI(\theta))$.

By the asymptotic normality theorem, the MLE attains the bound asymptotically: its asymptotic variance equals $1/(nI(\theta))$, so the MLE is **asymptotically efficient**.

One-parameter exponential family. A density of the form

$$f(x;\theta)=h(x)\exp(\eta(\theta)T(x)-A(\theta))$$

is a **one-parameter exponential family**. Its Fisher information is obtained by computing $I(\theta)=-\mathbb{E}_\theta[\partial^2/\partial\theta^2\log f(X;\theta)]$ from this form. Comparing $1/(nI(\theta))$ with the variance of a candidate unbiased estimator determines whether that estimator is efficient.

8 Asymptotic confidence intervals

Definition 16 (Asymptotic confidence interval) Let $z_{\alpha/2}$ be the upper $\alpha/2$ quantile of the standard normal distribution. From the asymptotic normality of the MLE, an **asymptotic confidence interval** of level $1-\alpha$ for θ is

$$\hat{\theta}_n \pm z_{\alpha/2}\sqrt{\frac{1}{nI(\theta)}}.$$

Definition 17 (Plug-in principle) Since θ is unknown, the asymptotic variance $1/(nI(\theta))$ is itself unknown. The **plug-in principle** replaces θ by its estimate $\hat{\theta}_n$, giving the usable interval

$$\hat{\theta}_n \pm z_{\alpha/2}\sqrt{\frac{1}{nI\left(\hat{\theta}_n\right)}}.$$

8.1 Sample size determination

The interval above has half-width

$$w = z_{\alpha/2}\sqrt{\frac{1}{nI(\theta)}}.$$

To guarantee a confidence interval of level $1-\alpha$ with half-width at most a specified w, solve this for n:

$$n \geq \frac{z_{\alpha/2}^2}{w^2\, I(\theta)}.$$

As $I(\theta)$ is unknown, one substitutes an estimate or a conservative value of $I(\theta)$ to obtain the minimum required sample size.